Group 1 - Transient and Frequency Response of Series RLC Circuit

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Due Mar 13, 10:00 PM

Objective

Design and analyze **series RLC circuit** to study and correlate:

- Transient response
- Frequency response

Tasks to be Achieved

Circuit Design

- Design both **series RLC** and **parallel RLC** circuits.
- Select suitable values of **R, L, and C** to realize:
- Underdamped condition
- Critically damped condition
- Overdamped condition

Transient Analysis (Time Domain)

- Obtain transient responses for all three damping conditions.
- Clearly identify and report:
- Presence or absence of oscillations
- Overshoot
- Settling behavior

Frequency Response Analysis

- Obtain the frequency response of each circuit.
- Identify and report:
- Resonant frequency
- Bandwidth
- Gain peaking (if any)

Observation of Key Phenomena

- Demonstrate:
- Resonance
- Peaking behavior
- Effect of resistance on damping and bandwidth

Correlation Study (Important)

- Establish a clear relationship between:
- Transient response (time domain)
- Frequency response (frequency domain)

Explain how:

- Underdamped response corresponds to peaking
- Overdamped response corresponds to no resonance
- Critical damping represents the boundary condition

Expected Outcome

- Establish a clear relationship between:
- Transient response (time domain)
- Frequency response (frequency domain)

Explain how:

- Underdamped response corresponds to peaking
- Overdamped response corresponds to no resonance
- Critical damping represents the boundary condition

Expected Outcome

Students should be able to:

- Relate **time-domain behavior** to **frequency-domain characteristics**
- Understand the role of **R, L, and C** in shaping system response
- Interpret damping in both domains consistently

Submission Requirements

- Circuit diagrams (series and parallel)
- Plots of transient responses
- Frequency response plots
- Comparative analysis of all three damping cases
- Clear explanation of correlation between time and frequency domains

You may take voltage across the capacitor to demonstrate the above findings.